Shubh Sehgal

Data Scientist

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SUMMARY

Data Scientist with a track record of shipping ML and NLP solutions from research to production. I specialize in building scalable data pipelines and MLOps workflows using Python, PyTorch, SQL and AWS. Beyond the code, I focus on translating complex model outputs into clear business strategies for stakeholders.

PROFESSIONAL EXPERIENCE

Research Data Scientist, **Computational Linguistics & Speech Processing Lab | Rochester, NY** **Aug 2025 - May 2026**

- Supported a $2M NSF-funded socially intelligent AI research program by building ML workflows for speech and Text analysis.
- Implemented a semantic search engine using OpenAI text-embedding-3-large embedding model and pgvector, indexing 200K+ articles with hybrid retrieval improving search relevance scores by 45% over keyword-only baseline.
- Architected a multi-agent research system using Claude Haiku 4.5 API and LangGraph with tool calling (web search, SQL, document retrieval) which resulted in cutting analyst research time from 4 hours to 25 minutes per report.
- Reduced Word Error Rate by 32% by developing scalable pipelines on RIT's on-premise GPU cluster using PySpark, PyTorch, and OpenAI Whisper to process 200+ hours of conversational audio.
- Reduced manual preprocessing effort by 25% by orchestrating Python and PySpark workflows with Apache Airflow for automated transcript cleaning, text normalization, and dataset preparation.
- Achieved 0.87 F1-score in text classification by fine-tuning DistilBERT, RoBERTa and DeBERTa models on labeled text datasets.
- Served models to production via FastAPI REST endpoints on the university's infrastructure, enabling real-time inference for research applications.

Data Scientist, **CUTSO LLP | Remote, India** **Dec 2023 - Aug 2024**

- Built an internal enterprise chatbot allowing SME stakeholders to query business data, integrated with Text-to-SQL capabilities for direct database interaction.
- Enhanced the chatbot through advanced chunking strategy experiments (recursive, semantic, parent-document), reducing hallucination rates from 18% to 4% and improving answer grounding by 3 times on internal evaluation benchmarks.
- Designed an anomaly detection framework for 10+ digital channels of RW, reducing monthly loss avoidance by $23K.
- Built an interactive dashboard for 120+ private sector clients to explore employment and investment opportunities across 30+ geographies.
- Developed a smart solution that reduced manual data entry errors from 89% to less than 5%, decreasing processing time from 20+ days to 48 minutes.

Associate Data Scientist, **actyv.ai | Bengaluru, India** **Jan 2023 - Sep 2023**

- Built and deployed an end-to-end XGBoost credit risk model on AWS using Docker to predict SME loan repayment risk, using DVC for data/model versioning and MLflow for experiment tracking improving ROC-AUC metric by 8%.
- Identified high-value SME customer segments by applying K-Means clustering to 60K transaction records with Pandas and Scikit-learn, helping marketing teams improve loan campaign targeting.
- Built a multimodal document classification system using Google Cloud Vision OCR, TF-IDF and a CNN + BERT ensemble model, achieving 0.95 precision with a Human-in-the-Loop workflow to route low-confidence predictions for manual review.
- Developed SQL based ETL pipelines on Amazon Redshift to clean, transform, and aggregate raw business data into ML-ready feature tables for downstream model training.
- Collaborated with cross-functional stakeholders to identify data-related bottlenecks, translate business requirements into analytical solutions, and support data-driven strategic decisions.

SKILLS

Languages & Databases : Python, C++, SQL, PostgreSQL, MySQL

Machine Learning & Statistics : Scikit-learn, PyTorch, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, PCA, Ensemble Methods

Data Analysis & Visualization : Pandas, NumPy, Matplotlib, Tableau, Power BI, Hypothesis Testing, A/B Testing

Data Engineering & MLOps : PySpark, Apache Airflow, Docker, Flask, FastAPI, MLflow, DVC, ONNX Runtime, Git, Hopsworks

Cloud & Infrastructure : Linux, AWS EC2, AWS SageMaker, AWS S3, AWS Lambda, Amazon Redshift, Azure File Storage, Azure OpenAI

Applied AI : Hugging Face, Transformers, RAG, Prompt Engineering, Fine-tuning, LangChain, LangGraph, LlamaIndex, FAISS, Vector DB (Pinecone, weaviate), LLM, Embeddings

PROJECTS

Clinically Supervised Mental Health Voice AI Agent

- Built a multi-agent Voice AI system using Claude API for university counseling with three AI agents: Screening (clinical decision trees), Social Media Context (real-time trend retrieval) and Summary & Handoff (clinician-ready reports with crisis detection guardrails) replacing static intake surveys.
- Currently in beta at RIT health services, saving counselors 15–20 minutes per session and delivering higher-quality screening data than traditional surveys.
- Created handoff documentation and trained counseling staff to review and act on AI-generated summaries independently.

Enterprise Compliance Chatbot

- Developed a Retrieval-Augmented Generation (RAG) chatbot to handle enterprise compliance queries across 4,000+ documents.
- Utilized hybrid retrieval methods including OpenAI embeddings, FAISS for vector search, BM25 for keyword search, and GPT-4o for response generation.
- Boosted retrieval quality and response speed by integrating metadata filtering, cross-encoder reranking, and semantic caching. These improvements significantly reduced irrelevant search results before generation.

Real-time AI-Powered Yoga Form Correction | Project Link

- Built a real-time computer vision application that classifies yoga poses and provides corrective posture feedback using pose estimation, supervised learning, and rule-based form validation.
- Developed a pose classification pipeline with MediaPipe and scikit-learn, extracting 33 skeletal landmarks, preprocessing landmark features, and training a Random Forest classifier for yoga pose recognition.
- Integrated Flask APIs and Gemini LLM to serve model predictions, generate personalized yoga recommendations, and deliver AI-assisted coaching through a responsive web interface.

End-to-End ML Pipeline for Seismic Magnitude Prediction | Github

- Built an end-to-end MLOps pipeline for seismic magnitude prediction, automating data ingestion, validation, feature engineering and model retraining using ANSS earthquake data.
- Developed an Airflow DAG on AWS EC2 to extract, transform, and load data from the ANSS API into S3, triggering Lambda functions for preprocessing.
- Used Great Expectations for automated feature validation, Hopsworks for feature storage and model versioning, and GitHub Actions for scheduled CI/CD workflows across training and inference pipelines.

EDUCATION

Rochester Institute of Technology - *MS, Artificial Intelligence* **Aug 2024 - May 2026**

NIIT University - *BTech, Computer Science* **Sep 2019 - Jun 2023**

ACHIEVEMENTS

Best Presentation Award, AWARE-AI Spring 2026 Hackathon **Mar 2026**

Won the Best Presentation Award for building a physiologically-aware multimodal AI chat interface integrating biosignal analysis and adaptive LLM interactions.
Certificate **:** https://drive.google.com/file/d/13f_ZCg1ewp5ivIjDRxskcc7G4pVBdcMe/view?usp=sharing

Academic Scholarship, Rochester Institute of technology **Jun 2025**

Recipient of an 80% tuition scholarship for academic and research performance.